Shivling Bhange

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Profile

Frontend Developer with **6 years** of experience building scalable, high-performance web applications for automotive and enterprise security domains. Proven track record in reducing report generation time by 60-65% for large datasets and improving application load times by 2x through performance optimization and efficient data parsing. Skilled in **React.js, Node.js, and TypeScript**, known for delivering robust, maintainable software through **TDD,BDD** practices and clean component architecture.

Skills

Frontend: **React.js, Redux,** React Hooks, Context API, Angular, **JavaScript (ES6+)**, **TypeScript**, HTML5, CSS3, MUI, Bootstrap, React Router, Context API, Axios, Webpack

Database: MongoDB

DevOps & Tools: Git, GitHub, Bitbucket, Jira, AWS (EC2, S3, Lambda), Docker Basics, Postman

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Testing: Jest, React Testing Library (**RTL**), Mocha, Chai, BDD (Gherkin), **TDD**, Unit Testing, Integration Testing

Other: CI/ CD, Agile/ Scrum, Unit Testing (Jest, Mocha)

Professional Experience

Software Engineer – – BMW Techworks India (**BMW Group**) *01/2024 – Present | Pune, India*

- **Tech Stack** : React.js, Redux Toolkit, Node.js, Express.js, MongoDB, Git Actions.
- Improved system performance by reducing report generation time by 60-65% for 150-200MB datasets through memoization and pagination techniques, enabling faster data processing for engineering teams.
- **Architected and maintained a production-grade component library** using React, TypeScript, and CSS-in-JS / design tokens, serving as the single source of truth for all customer-facing and internal web apps.
- **Pioneered an AI-driven design workflow** leveraging Claude (for UI generation and layout exploration) and Cursor (for rapid development) to independently bridge design gaps and scale UI consistency without a dedicated designer.
- Developed a high-precision ECU comparison tool using data structure & algorithms to optimize rendering pipelines, applying analytical precision to debugging complex parameter changes for 200+ engineers, reducing analysis time for 200K+ parameters to under 5 seconds.
- Implemented Role-Based Access Control (RBAC) with Admin and Application Developer roles for 200+ users, ensuring secure access and user-level permissions in compliance with ASPICE and ISO standards.
- Built advanced UI tools including search, filters, and visual signal comparisons using React.js and WASM, improving workflow efficiency by 70% for 200+ engineers.
- Automated report generation, reducing manual review effort for ECU calibration validation improved accuracy and speed.
- Built a reusable React component library with 500+ unit and integration tests using Jest, React Testing Library, and Mocha following TDD/BDD practices, achieving 90% code coverage and significantly reducing regression bugs across the application.

Software Engineer – – Colortokens Inc *06/2022 – 01/2024* | *Banglore, IND*

- **Tech Stack**: React.js, Angular 18 , Node.js, MongoDB, Express.js, EC2, S3 .
- Designed a centralized admin dashboard to manage and monitor 100K+ endpoints across the network, displaying hostname, IP, groups, firewall rules, policies, and system status, enabling real-time visibility and reducing manual monitoring effort.
- Provisioned and configured **AWS EC2** instances to host and deploy web applications.
- Enabled administrators to view and manage firewall rules for all 100K+ onboarded systems from a centralized interface, eliminating the need to check each system individually and ensuring consistent network traffic control and unauthorized access blocking.
- Optimized REST API performance through indexing and query optimization using NodeJS, reducing initial load time for 200K+ records from 6 seconds to 3 seconds, achieving 2x faster data retrieval.
- Implemented remote resource decommissioning to securely offboard endpoints, automatically deleting system policies, access tokens, and associated data in the background, ensuring proper offboarding and compliance.
- Developed audit logs, alerts, role-based permissions, and alert history for security compliance, reducing audit log load time by 70% through API optimization.
- Provided real-time visibility into endpoint health, connection status, OS-level details, and threat alerts.

Software Engineer – GravityTech Software Solution *01/2021 – 06/2022* | *Pune, IND*

- Engineered secure and scalable internal web applications for a banking institution, ensuring compliance with industry-grade security protocols and data protection standards.
- Developed global access portals serving internal clients across multiple regions, enabling seamless cross-location collaboration and centralized access to critical banking resources.
- Delivered high-performance applications while ensuring strict adherence to internal security protocols and banking industry compliance standards.
- Implemented Multi-Factor Authentication (MFA) to enhance security during user registration and account creation in web and mobile applications, reducing unauthorized access risks.
- Created intuitive, data-driven UI dashboards that improved internal operations and enabled faster, more informed decision-making for banking teams.
- Designed and implemented front-end solutions that streamlined workflows and enhanced user experience, driving measurable improvements in operational efficiency.

Education

SGGS Institute of Engineering & Technology (Gov. Autonomous) *06/2019*

B.Tech. (Electronics & Telecommunication)

Certificates

Test-Driven Development for JavaScript

Introduction to Agile Development and Scrum